

Karan Surana

Software Engineer — 044 807 4273 — karansurana23@gmail.com

Profile

Experienced Software Engineer with ~3 years of hands-on experience building scalable, production grade web applications. Adept at **system design and architecture**, translating complex requirements into clean, maintainable, and scalable technical solutions. Proficient in crafting responsive UIs and robust backend systems, with practical exposure to cloud infrastructure and modern DevOps practices.

Education

UNSW Sydney

Masters in Information Technology

Majors: Artificial Intelligence — WAM: Distinction

Sep 2022 - Sep 2024

Sydney

Key Skills

Frontend Technologies: React, TypeScript, JavaScript, HTML, CSS

Backend Technologies: NextJS, NodeJS, ExpressJS, RESTful APIs

DevOps and Tools: Jira, Confluence, Git, Docker, Kubernetes, AWS (EC2, S3, Lambda, Amplify), Kafka, CI/CD

Additional Skills: Redis, MySQL, MongoDB, PostgreSQL, Data Structures & Algorithm, Unit & Integration Testing.

Professional Experience

youXpowered

Mid Level Software Engineer

January 2025 - Present

Sydney, Australia

- **Architected and led** a team of 8 in the development of youXinsights, an enterprise grade, in-house analytics and visualization platform that replaced Power BI, serving 10+ organisations.
- Collaborated with **stakeholders** to transform business requirements into intuitive, dynamic reports and dashboards tailored to company specific workflows.
- Engineered **real-time data** visualization infrastructure with 60 second refresh intervals, implementing advanced cross-filtering and interactive dashboards across **25+ report types**.
- **Solely owned the complete lifecycle** (design, development, deployment, and support) of **youXapply**, another product, matching customers with lenders, significantly enhancing customer loan conversions.
- Acted as the **primary troubleshooter** for all product related issues, handling bug fixes, performance optimizations, and feature rollouts.
- Architected and developed an **AI based microservice** that auto generates lender notes for brokers transforming raw application data into concise narratives and reducing manual write time by **90%**.
- **Tech Stack:** TypeScript, JavaScript, NextJS, React, NodeJs, AWS.

Gida Technologies

Full Stack Developer

January 2021 - August 2022

- Improved backend development and API integrations for the Antara project, reducing API response time and computational overhead by 60% within the given timeframe.
- Assisted in frontend development, optimizing API calls and reducing frontend API response time by 40%, improving page load times and user interaction speeds.
- Implemented the complete inventory management system by designing the backend architecture and optimizing database queries through indexing and query optimization techniques.
- Implemented caching mechanisms using Redis to reduce server load and response times.
- Integrated API endpoints with multi-threaded handling for efficient parallel processing of large-scale customer orders, improving system performance by 30%.
- **Tech Stack:** TypeScript, React, Kafka, AWS, Redis, Zoho, Shopify, PostgreSQL.

Major Projects

Official Computer Booking System for VR Lab (UNSW) [Click here to access the Hosted Web App]

- Developed a full-stack web application that transformed the booking process for advanced computational workstations in the VR Lab.
- Implemented secure user authentication and dynamic booking management, enabling users to effortlessly reserve, view, and manage their computer bookings.
- Engineered seamless integration between frontend, backend, and database systems, ensuring efficient communication and robust performance across the platform.
- Enhanced user experience with intuitive interfaces and real-time updates, contributing to increased utilization and accessibility of lab resources.
- Contributed to a more efficient allocation of lab resources, reducing booking conflicts and wait times by 45%.

Asset Finance Management Platform [Click here to access the Hosted Web App]

- Developed a single-page application for managing finance applications for loans, using the MERN stack.
- Designed a responsive and scalable frontend enabling users to create, update, and delete finance applications, capturing detailed financial data.
- Built a RESTful API with Node.js and Express for backend logic, integrating with a MongoDB serverless cluster.
- Implemented logging, monitoring, and CI/CD pipelines for seamless deployments and error tracking.
- Deployed the application on AWS, incorporating serverless functions for efficient backend operations.

References

Nathan O'Donnell - youXpowered

Ali Darejeh - UNSW Sydney

Ayush Shukla - Gida Technologies

Links

 [KaranSurana](#)  [KaranSurana](#)
<https://karansurana.github.io/>